

Class 09: Candy Mini Project

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Background

Today we are taking a small detour to analyze a fun dataset that we have more intrinsic insight into. We will be using principal component analysis. Building on your work with PCA in the previous labs (UK Foods, RNA-seq and breast cancer biopsy analysis), this mini-project gives you the opportunity to apply your data analysis skills to a new, tastier dataset.

Importing candy data

```
candy_file <- "candy-data.txt"

candy = read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1			0	0		1
3 Musketeers	1	0	0			0	1		0

One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

The dataset includes all sorts of information about different kinds of candy. For example, is a candy chocolaty? Does it have nougat? How does its cost compare to other candies? How many people prefer one candy over another?

Q1: How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2: How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

One of the most interesting variables in the dataset is **winpercent**. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

Q3: What is your favorite candy (other than Twix) in the dataset and what is its **winpercent** value?

```
candy["3 Musketeers", ]$winpercent
```

```
[1] 67.60294
```

Q4: What is the winpercent value for “Kit Kat”?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>  
  filter(row.names(candy)=="Kit Kat") |>  
  select(winpercent)
```

```
      winpercent  
Kit Kat    76.7686
```

Q5: What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Side-note: the `skimr::skim()` function

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let’s install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6: Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

`winpercent` is on a different scale; it has a mean of 50 and a standard deviation of 15, while the rest of the columns are between 0 and 1.

Q7: What do you think a zero and one represent for the `candy$chocolate` column?

A zero represents not containing chocolate, a one represents containing chocolate.

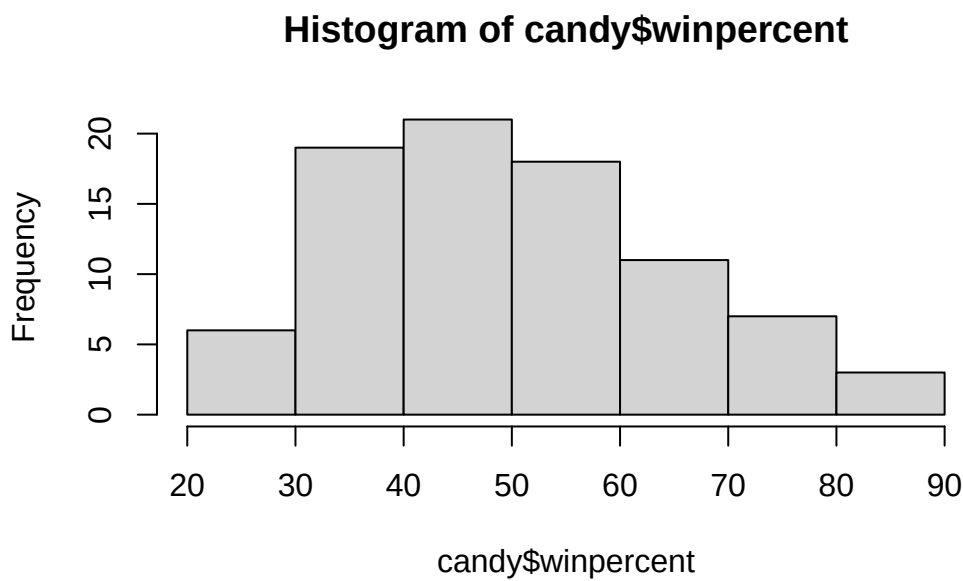
Exploratory analysis

A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`. Either works well in this case and (as always) it's your choice.

Q8: Plot a histogram of `winpercent` values using both base R and `ggplot2`.

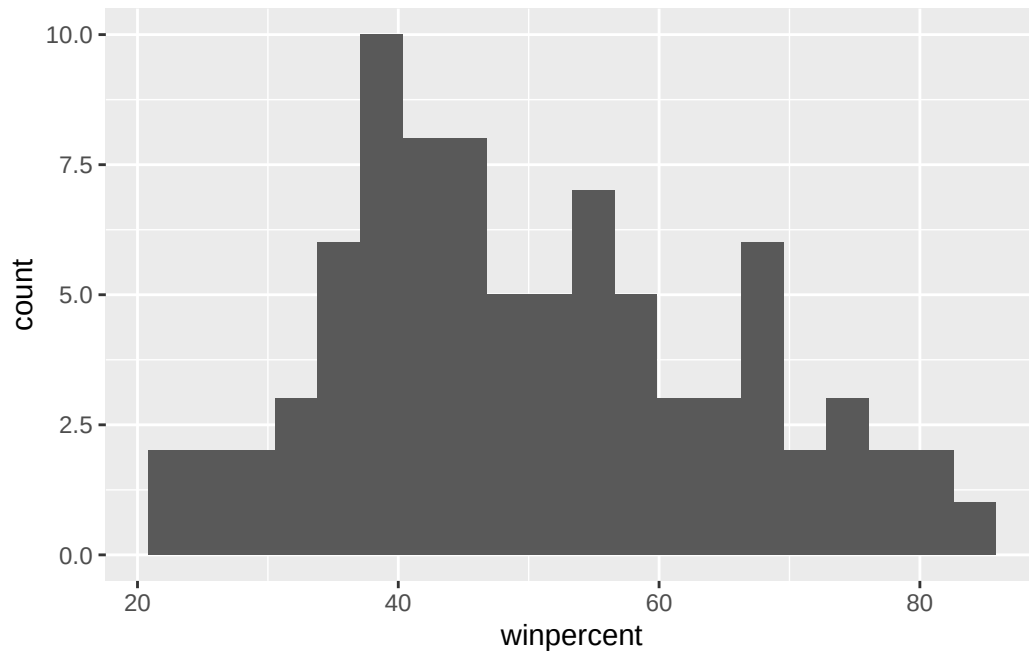
Base R

```
hist(candy$winpercent)
```



ggplot2

```
library("ggplot2")  
  
ggplot(candy, aes(x = winpercent)) + geom_histogram(bins = 20)
```



Q9: Is the distribution of winpercent values symmetrical?

The distribution is not symmetrical, there seems to be a higher concentration of winpercent values near 40 percent, skewed.

Q10: Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The median is a better measure of centrality, and it is below 50%.

Q11: On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Chocolate is higher ranked on average.

Q12: Is this difference statistically significant?

```
t.test(candy$winpercent[as.logical(candy$chocolate)], candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

It is significant, the p value is < 0.05 .

Overall Candy Rankings

Let's use the base R `order()` function together with `head()` to sort the whole dataset by a variable/column of interest. Or if you have been getting into the tidyverse and the dplyr package you can use the `arrange()` function together with `head()` to do the same thing and answer the following questions:

Q13: What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent) |>
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14: What are the top 5 all time favorite candy types out of this set?

```
candy |>
  arrange(desc(winpercent)) |>
  head(5)
```

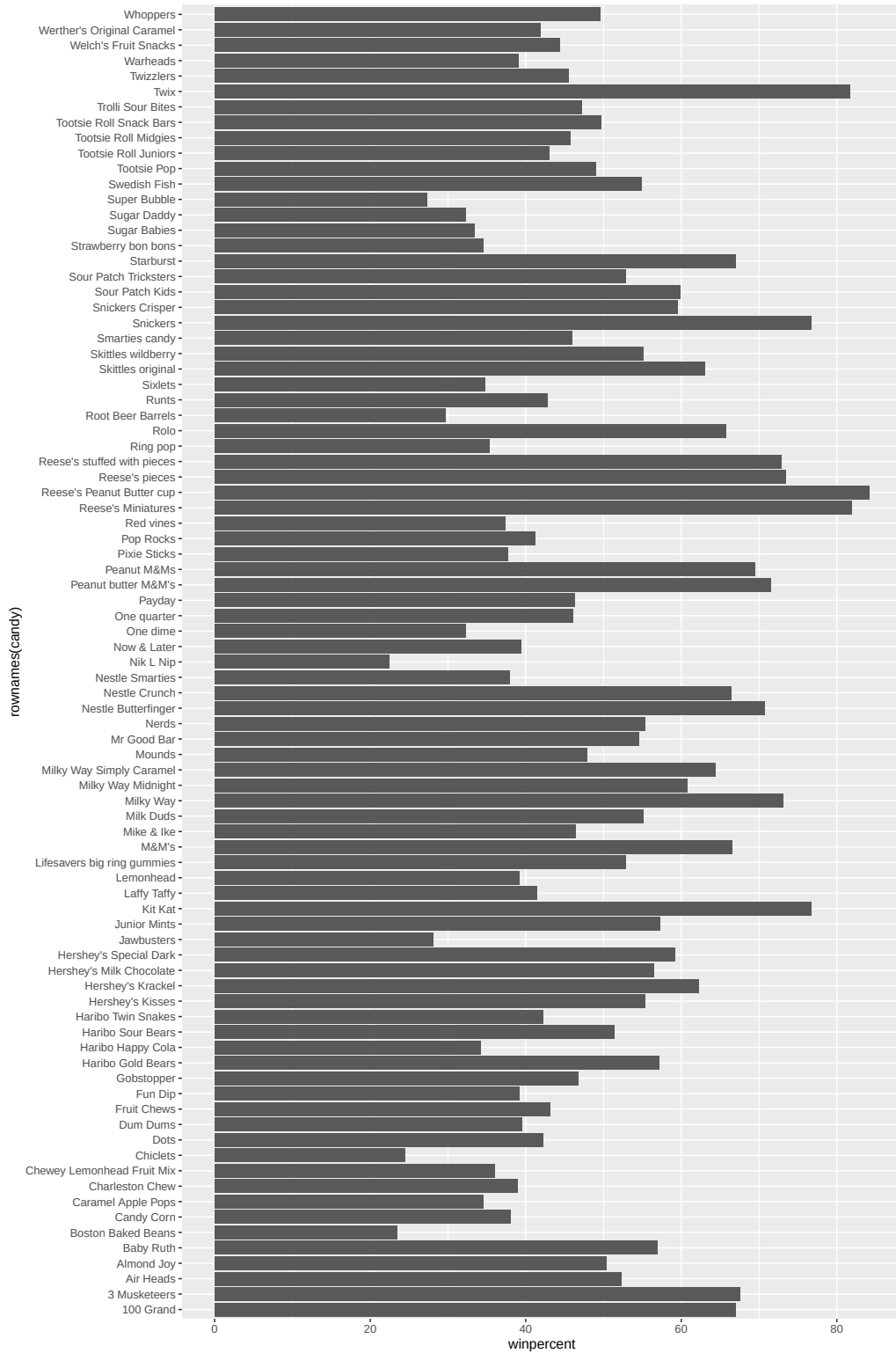
	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup				0	0	0	0	0.720
Reese's Miniatures				0	0	0	0	0.034
Twix				1	0	1	0	0.546
Kit Kat				1	0	1	0	0.313
Snickers				0	0	1	0	0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

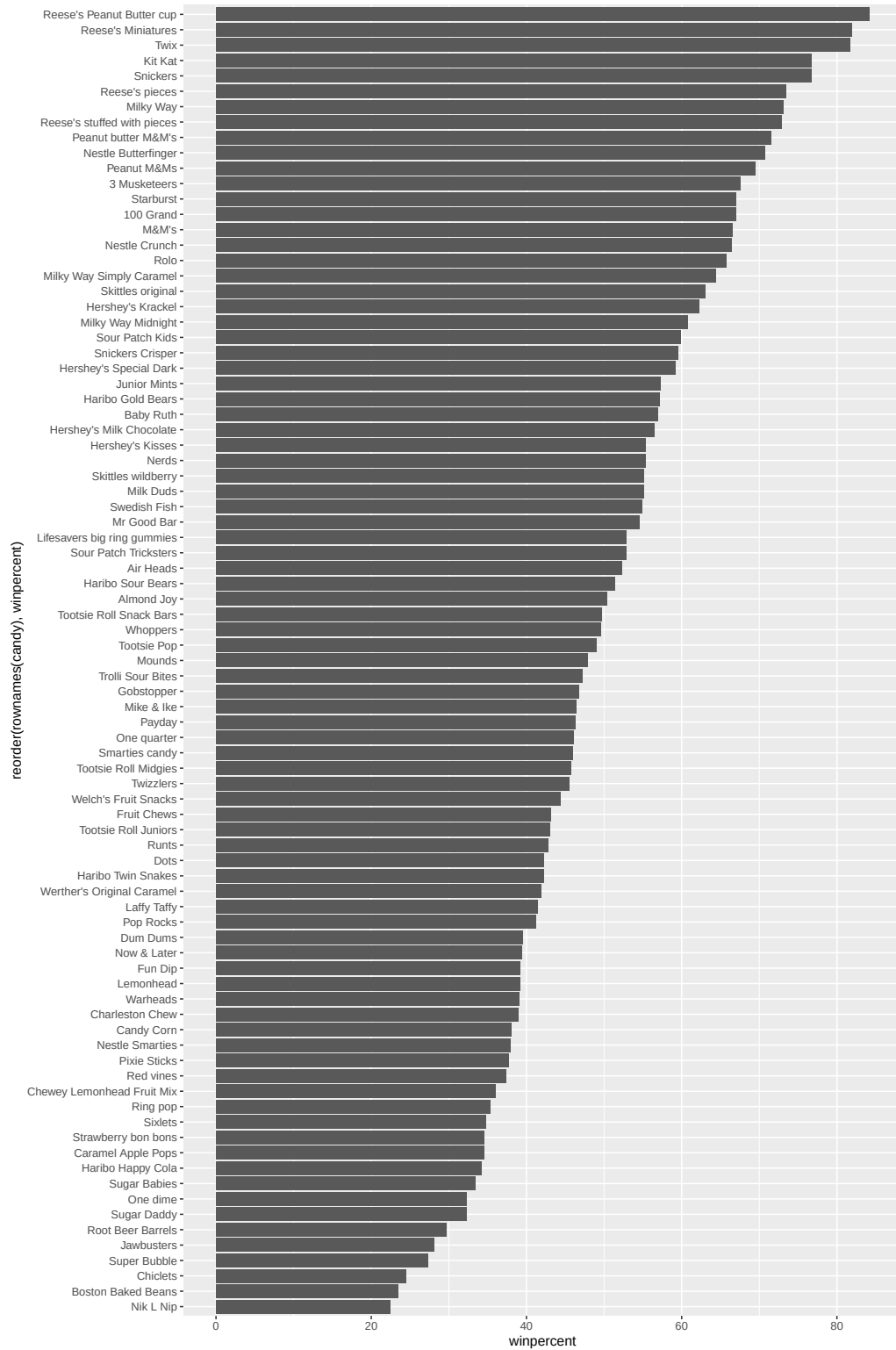
Q15: Make a first barplot of candy ranking based on `winpercent` values

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



Q16: This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



Time to add some useful color

Let's setup a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
my_cols=rep("black", nrow(candy))

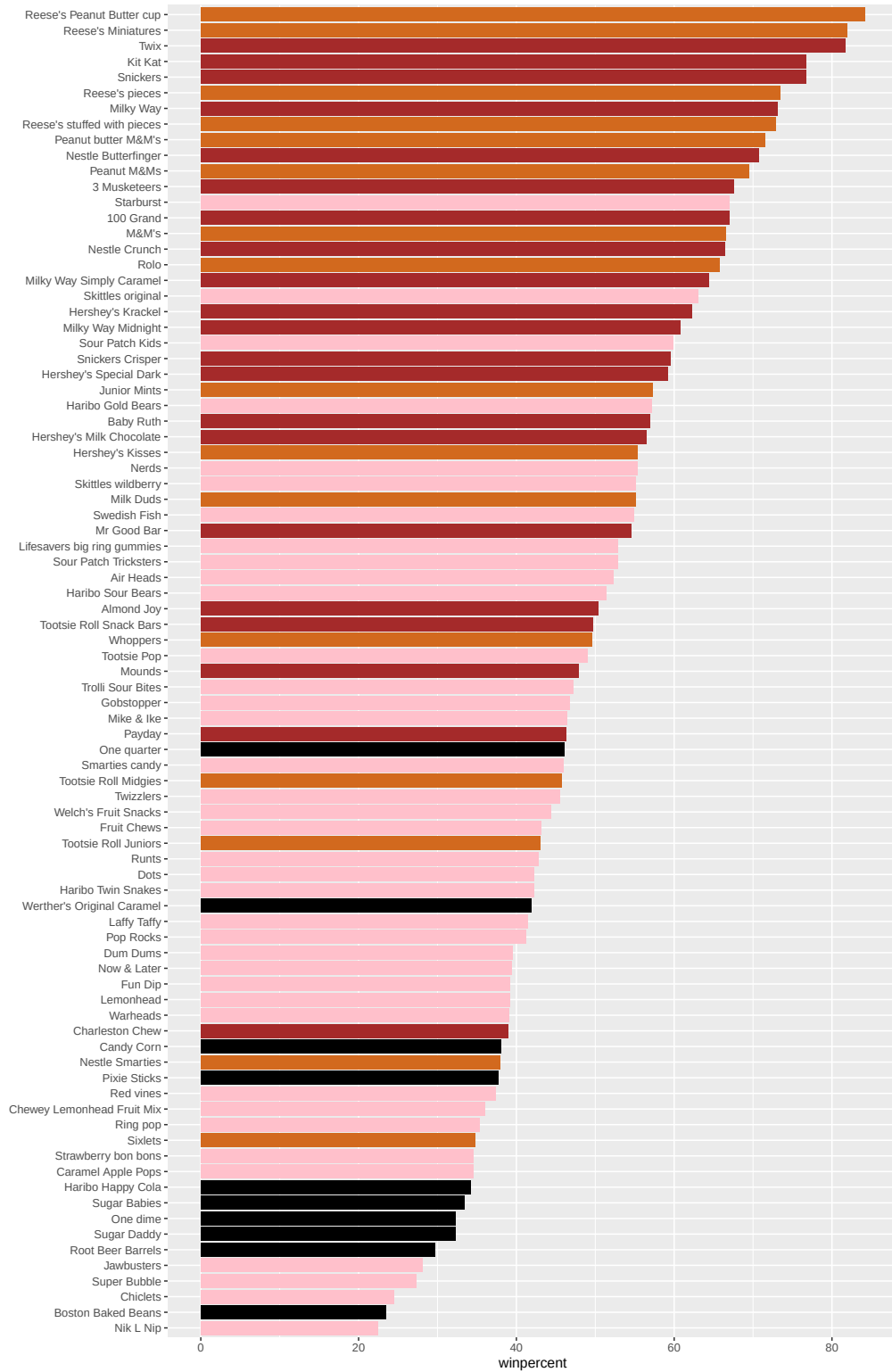
my_cols[as.logical(candy$chocolate)] = "chocolate"

my_cols[as.logical(candy$bar)] = "brown"

my_cols[as.logical(candy$fruity)] = "pink"
```

Now let's try our barplot with these colors. Note that we use `fill=my_cols` for `geom_col()`. Experiment to see what happens if you use `col=mycols`.

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17: What is the worst ranked chocolate candy?

Sixlets

Q18: What is the best ranked fruity candy?

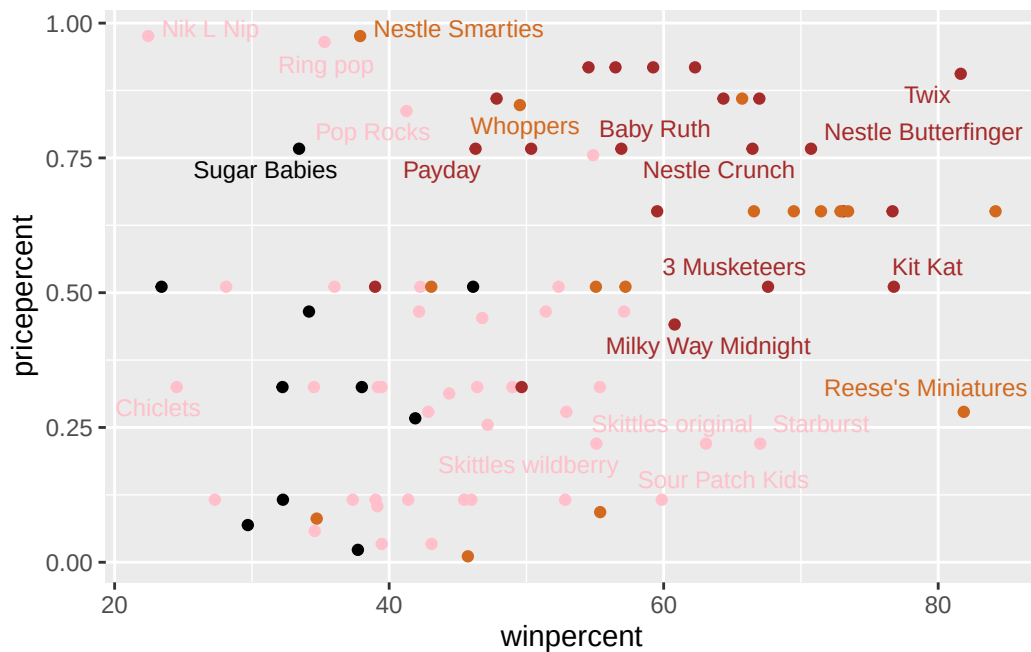
Starburst

Taking a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of `winpercent` vs the `pricepercent` variable. The `pricepercent` variable records the percentile rank of the candy's price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(x=winpercent, y=pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19: Which candy type is the highest ranked in terms of `winpercent` for the least money - i.e. offers the most bang for your buck?

```
candy$value = candy$winpercent / candy$pricepercent
candy[which.max(candy$value), ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Tootsie Roll Midgies	1	0	0	0	0
	crispedricewafer	hard bar	pluribus	sugarpercent	
Tootsie Roll Midgies	0	0	0	1	0.174
	pricepercent	winpercent	value		
Tootsie Roll Midgies	0.011	45.73675	4157.886		

Q20: What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
candy[order(candy$pricepercent, decreasing = TRUE),] [1:5,]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Nestle Smarties	1	0	0	0	0
Ring pop	0	1	0	0	0
Hershey's Krackel	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
	crispedricewafer	hard bar	pluribus	sugarpercent	
Nik L Nip	0	0	0	1	0.197
Nestle Smarties	0	0	0	1	0.267
Ring pop	0	1	0	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Hershey's Milk Chocolate	0	0	1	0	0.430
	pricepercent	winpercent	value		
Nik L Nip	0.976	22.44534	22.99728		
Nestle Smarties	0.976	37.88719	38.81884		
Ring pop	0.965	35.29076	36.57073		
Hershey's Krackel	0.918	62.28448	67.84802		
Hershey's Milk Chocolate	0.918	56.49050	61.53649		

Nik L Nip is the least popular

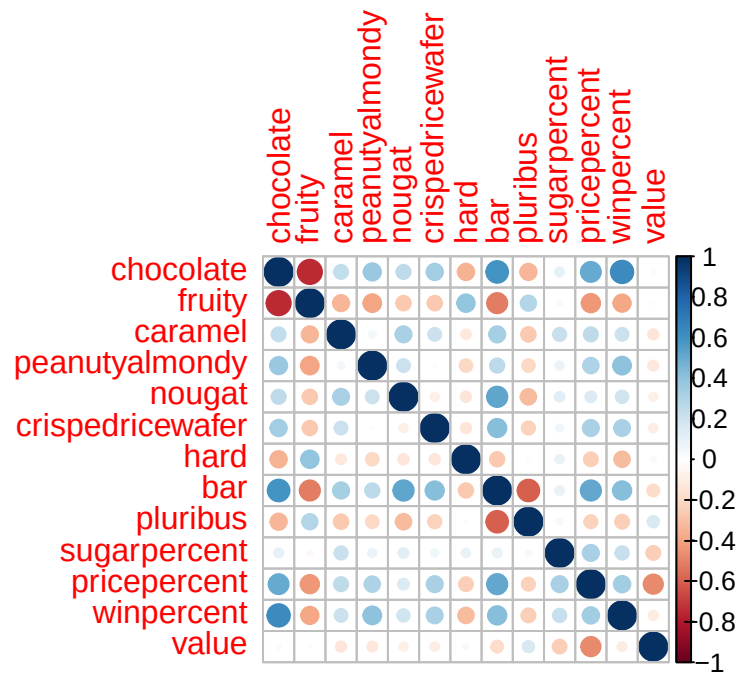
Exploring the correlation structure

We can calculate the pair-wise correlation of all our columns

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)  
corrplot(cij)
```



Q22: Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are strongly anti-correlated. Bar and pluribus are also anti correlated.

Q23: Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Chocolate and fruity will contribute most to PC1 because they are strongly negatively correlated and create a major split in the data. Winpercent and pricepercent also contribute since they have larger variation and help capture differences in popularity and cost. Together, these variables drive the main separation between candies.

Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` because we have one variable `winpercent` that is on an very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

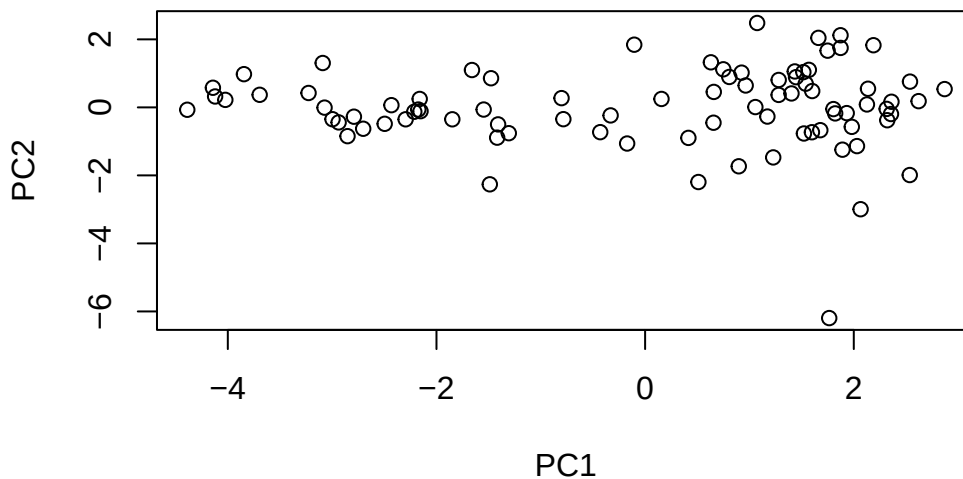
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0938	1.2127	1.13054	1.0787	0.98027	0.93656	0.81530
Proportion of Variance	0.3372	0.1131	0.09832	0.0895	0.07392	0.06747	0.05113
Cumulative Proportion	0.3372	0.4503	0.54866	0.6382	0.71208	0.77956	0.83069

	PC8	PC9	PC10	PC11	PC12	PC13
Standard deviation	0.78462	0.68466	0.66328	0.57829	0.43128	0.39534
Proportion of Variance	0.04736	0.03606	0.03384	0.02572	0.01431	0.01202
Cumulative Proportion	0.87804	0.91410	0.94794	0.97367	0.98798	1.00000

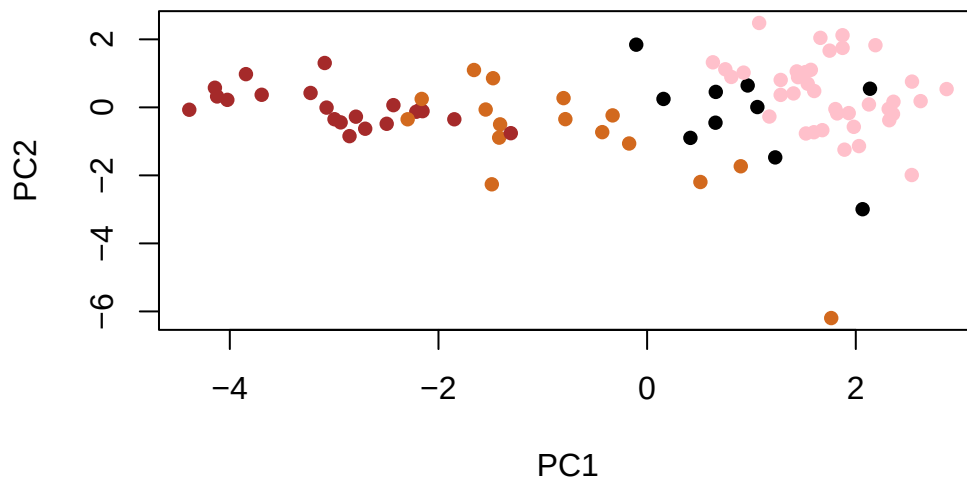
Now we can plot our main PCA score plot of PC1 vs PC2.

```
plot(pca$x[,1:2])
```



We can change the plotting character and add some color:

```
plot(pca$x[,1:2], col=my_cols, pch=16)
```

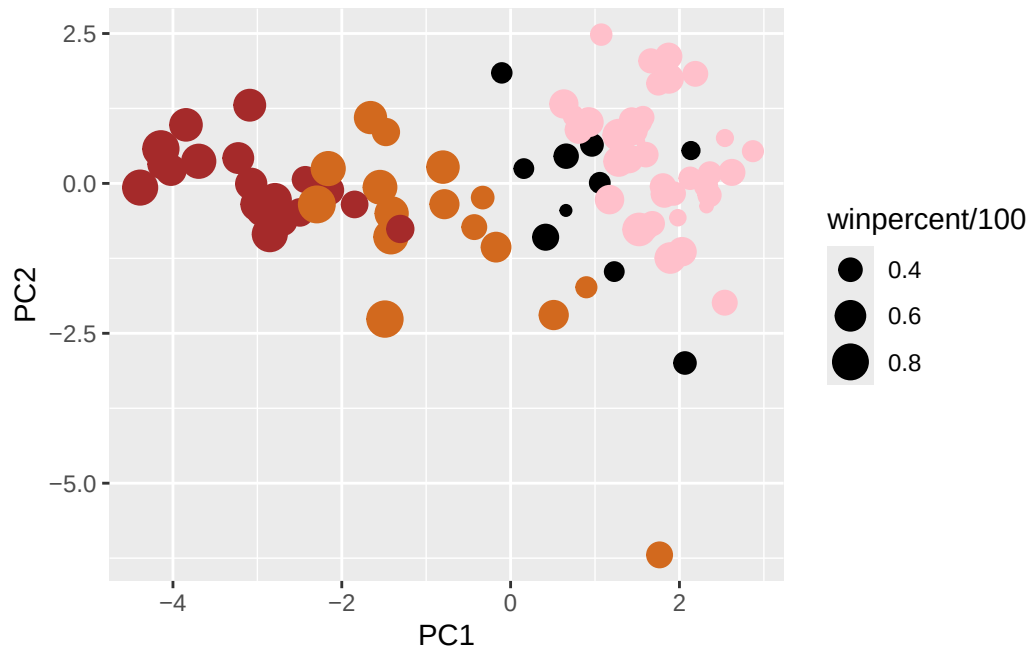


We can make a much nicer plot with the ggplot2 package but it is important to note that ggplot works best when you supply an input data.frame that includes a separate column for each of the aesthetics you would like displayed in your final plot.

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)

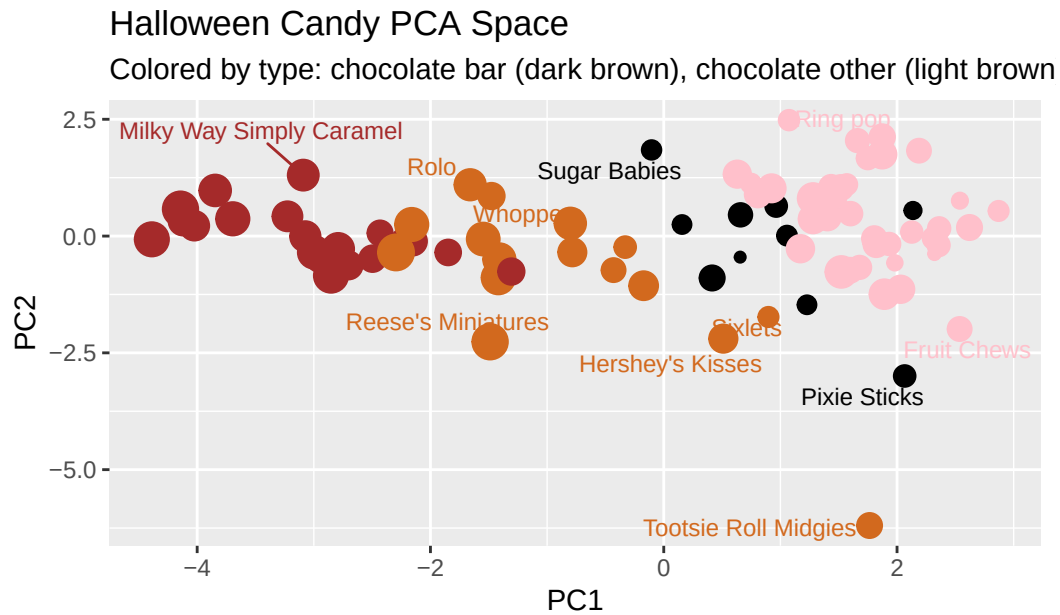
p
```



Again we can use the `ggrepel` package and the function `ggrepel::geom_text_repel()` to label up the plot with non overlapping candy names like. We will also add a title and subtitle like so:

```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```



If you want to see more candy labels you can change the `max.overlaps` value to allow more overlapping labels or pass the `ggplot` object `p` to `plotly` like so to generate an interactive plot that you can mouse over to see labels:

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

`last_plot`

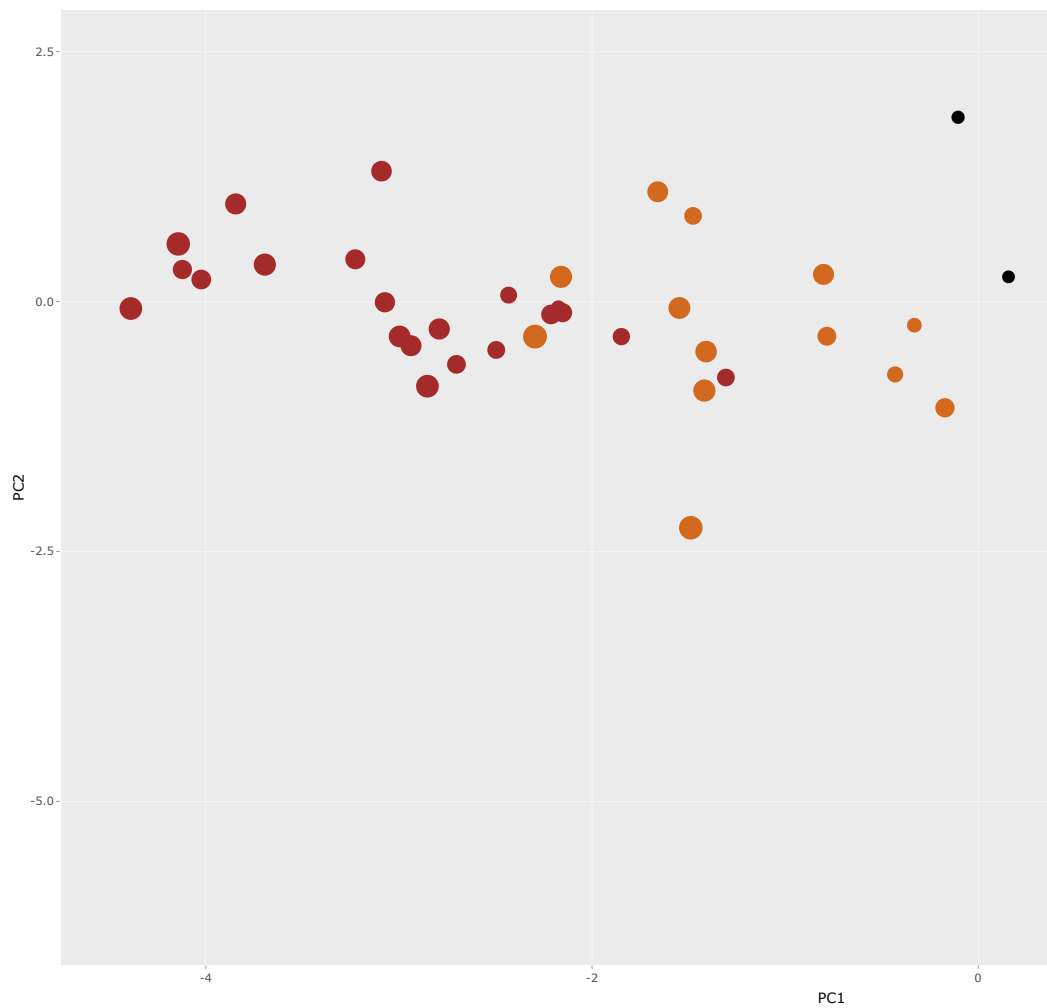
The following object is masked from 'package:stats':

`filter`

The following object is masked from 'package:graphics':

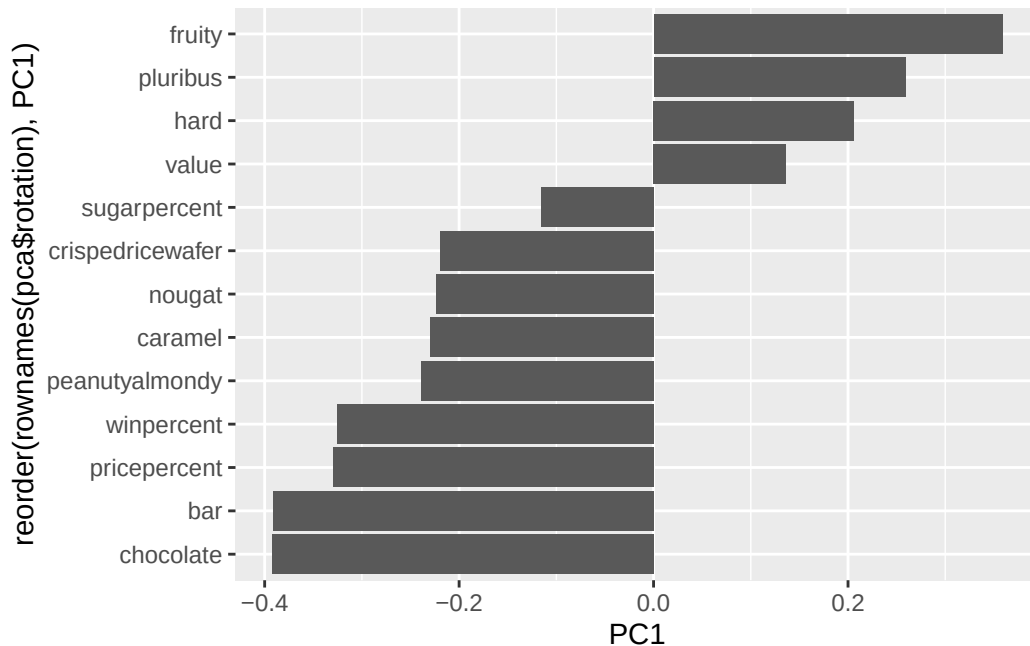
`layout`

```
ggplotly(p)
```



Let's finish by taking a quick look at PCA our loadings. Do these make sense to you? Notice the opposite effects of **chocolate** and **fruity** and the similar effects of **chocolate** and **bar** (i.e. we already know they are correlated).

```
ggplot(pca$rotation) +  
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +  
  geom_col()
```



Q24: Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables picked up strongly in the positive direction of PC1 are **fruity**, **pluribus**, and **hard**. These make sense because they represent a group of non-chocolate, often bagged or hard candies, which contrasts with chocolate-based candies.

We previously saw this relationship in the correlation plot, where **fruity** and **chocolate** were strongly negatively correlated, showing the same separation that PC1 is capturing.

Summary

The PCA loadings revealed that the primary axis of variation separates chocolate-based candies from **fruity** ones - the same pattern we observed in the correlation matrix, now visualized in a single, interpretable plot.

Q25: Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy is typically chocolate-based, often a bar, and not **fruity**, with higher winpercent and reasonable price. Visualizations showed chocolate candies clustering at higher popularity, while the correlation plot highlighted the strong negative relationship between **chocolate** and **fruity**. PCA confirmed this pattern by separating candies along PC1 based on **chocolate** vs **fruity**, reinforcing that candy type is the main driver of preference.